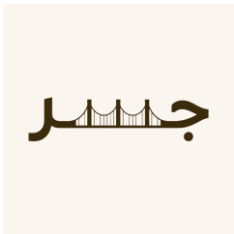


LIBYA ARTIFICIAL INTELLIGENCE FORUM (Ai4LY 2026) • NATIONAL CODATHON

JISR (جسر) — BRIDGE

NOTE



AI-Assisted Writing Companion for Young People in Libya

A privacy-first, Arabic-native authoring companion engineered to break the silence at the 'First Sentence Problem' by turning everyday stress chips and raw thoughts into 3 tone-adapted message drafts for trusted human contacts.

Project Dimension	Specification & Repository Attribute
Event & Track	Ai4LY National Codathon 2026 — Track: AI for Mental Health & Youth Well-Being
Team Roster	Mohamed Abdel Wadod Thabet (Team Lead & Trust Lead), Rayan Khalid Aljabo (Mobile), Muetazballah Qambar (AI/Backend), Shaima Abdulsalam Aljelali (Safety)
Submission Channel	GitHub Project Repository (Active Branch: main) with Standalone Android APK & Web Showcase
Submission Deadline	Wednesday, October 7, 2026, 11:50 PM (Presentation: Thursday, October 8, 2026)
Client Platforms	React Native 0.74 / Expo SDK 51 (com.ai4ly.jisr) • Next.js 16 Web Showcase • Standalone APK
AI Backend Engine	Stateless Python FastAPI Microservice • Gemini 1.5 Flash (sub-800ms) • Groq Llama 3.3 70B Failover
Verified Safety Gate	Guardian Layer: 100.0% Crisis Recall (10/10 caught, 95% CI 72.2%–100%) • 450 Passing Unit Tests
Privacy Model	Zero Accounts • Zero Cloud Retention • Pre-Transmission PII Scrubbing (+218/emails/names)

[*] EXECUTIVE STATUS: FULL CODATHON COMPLIANCE VERIFIED

All official deliverables required by the Ai4LY National Codathon are 100% complete, verified, and committed in the repo: Scope A (Working Mobile Client & Next.js 16 Web Showcase), Sideloadable Android APK (builds/jisr-v1.0.0.apk, 63.1 MB), Scope B (Documentation Hub & README.md), Scope C (Technical Report REPORT.md with Wilson 95% CIs), Scope D (Pitch Deck PITCH_DECK.pptx & Defense Q&A), Section 4 Citations Register, and 450 automated tests (npm test pass).

1. Executive Summary & Core Mission

In moments of acute emotional distress, the single greatest point of failure is silence. For young people in Libya navigating everyday academic pressures, family expectations, employment uncertainty, interpersonal strain, and financial stress, the barrier to seeking support is rarely an absence of people who care. Rather, it is the overwhelming friction of composing the very first sentence.

Jisr (Bridge Note) is a lightweight, Arabic-native, privacy-first mobile companion engineered specifically to bridge this gap. Operating on minimal user friction—one-tap selection of situation chips and an optional 1 to 3 lines of messy, unstructured text—Jisr instantly restructures raw emotions into three tone-adapted, culturally authentic message drafts (Gentle, Direct, Formal) addressed to a real person in the user's circle (a friend, sibling, parent, trusted elder, or teacher).

[-] PRODUCT PHILOSOPHY: HUMAN-TO-HUMAN BRIDGE, NOT CHATBOT

CORE PARADIGM: Jisr is strictly NOT a chatbot, NOT an artificial therapist, and NEVER engages in open-ended conversations. Its entire purpose is to help the user write one message, hand it off directly to WhatsApp, Messenger, or SMS via the native OS share sheet, and exit the application within 60 seconds. The metric of success is not screen engagement, but a real-world human conversation started.

2. Problem Definition & The Libyan Realities

2.1 The Triad of Silence at the First Sentence

When an individual experiences escalating stress, three psychological and cultural forces compound to enforce silence:

- **1. Cultural Stigma & Fear of Labeling:** Mental health struggles have historically been guarded as private family matters. Naming one's pain out loud using clinical labels carries severe social and familial stigma.
- **2. Emotional Overwhelm:** Acute distress severely constrains working memory and executive cognitive bandwidth. The individual knows they are suffering but cannot organize their thoughts into a coherent message.
- **3. Articulation Energy Cost:** Converting internal anguish into an actionable text that invites comfort without triggering panic or sounding pathetic requires high emotional energy—energy an overwhelmed youth lacks.

2.2 The Scarcity of Specialist Care vs. Mobilizing Community Pillars

Specialized psychiatric and clinical psychological care in Libya is geographically concentrated (primarily in major hospitals in Tripoli and Benghazi), heavily overburdened, and frequently inaccessible or socially intimidating. Clinical infrastructure cannot—and should not—be the primary frontline response for everyday exam anxiety, interpersonal conflict, or routine situational distress. The most effective, immediate, and enduring support already

exists within the young person's immediate social fabric: an older brother, a close university friend, an empathetic mother, or a trusted mentor. Jisr activates these natural human safety nets by removing the initial outreach barrier.

2.3 Infrastructure & Socio-Technical Constraints in Libya

Any technological intervention deployed in Libya must reckon with local operational realities:

- **Infrastructure Resilience:** Periodic electrical load shedding and spotty 3G/4G connectivity demand an architecture that works offline without cloud reliance.
- **Hardware Footprint:** Many young people utilize budget Android devices; bloated apps or multi-gigabyte models cause severe thermal throttling and device freezing.
- **Messaging Ubiquity:** Meta Messenger and WhatsApp account for over 90% of interpersonal communication in Libya. Solutions must integrate seamlessly into these platforms.
- **Linguistic Calibration:** Young Libyans communicate in Libyan Arabic (اللهجة الليبية البيضاء), Modern Standard Arabic (MSA), and Latin-script Arabizi. Models must recognize and respect all three.

3. System Architecture: The 4-Layer Engine

Jisr's architecture decouples user capture, deterministic safety verification, AI drafting, and user trust into four resilient layers:

Layer	Technical Component	Core Functionality & Defensive Safeguard
Layer 1: Capture Layer	CaptureScreen.tsx (React Native / Next.js)	• 7 One-Tap Stress Chips: Exams (الامتحانات), Family (العائلة), Work (العمل), Relationships (العلاقات), Sleep (النوم), Money (المال), Other (أخرى). • 5 Recipient Selectors: Friend (صديق), Sibling (أخ/أخت), Parent (أحد الوالدين), Trusted Elder (شخص كبير), Teacher/Mentor (مرشد). • 1–3 Lines Free Text Box (Optional): Zero typing required if user is exhausted.
Layer 2: Guardian Safety Layer	crisisCheck.ts / evaluate.mjs safety/crisis-phrases.json safety/forbidden-terms.json	• Stage 1 (Pre-Draft Screener): 60 Libyan dialect distress phrases across 360+ spelling variants, with 7 whitelisted benign idioms (e.g., 'نموت من الضحك'). • Stage 2 (Static Support Card): Halts drafting immediately on crisis trigger; shows unalterable Emergency Room guidance. • Stage 3 (Clinical Blacklist): Blocks 55 clinical diagnostic and medication terms post-generation.
Layer 3: Drafting Engine (AI Core)	FastAPI Microservice & Local Offline Templates (Gemini 1.5 Flash / Groq)	• Online Path: Asynchronous FastAPI service returning 3 tone-adapted drafts (Gentle, Direct, Formal) in <800ms via Gemini 1.5 Flash (Groq Llama 3.3 70B failover). • Offline Path: 35 pre-vetted deterministic template permutations (plain-templates.json) triggered if network is down or API times out (2.5s).
Layer 4:	BaselineComparison.tsx	• Baseline Comparison: Live toggle comparing AI draft vs. static

Trust, Privacy &
Native Sharing

FaithfulnessView.tsx
OutboundPreview.tsx
Share.share()

template to prove real AI value.

- Faithfulness View: Word-level visual attribution mapping generated sentences back to user input.
- Outbound Preview: Transparency inspector showing scrubbed PII before API transit.
- Native Share Sheet: Direct hand-off to WhatsApp/SMS with zero cloud telemetry.

3.1 Visual Interface & Mobile User Journey

The mobile user experience is designed for maximum calm and minimal cognitive friction. Below is the mobile interface showcase illustrating the RTL situation chips, recipient selector, and persistent human route:



Figure 1: Jisr Mobile Client Interface (com.ai4ly.jisr) — Zero-Pressure Arabic RTL Capture Flow

4. The Guardian Safety Architecture & Ethical Contacts Policy

4.1 Multi-Stage Defensive Perimeter

Safety is not treated as an afterthought or a system prompt suggestion; it is enforced deterministically by code:

- **1. Pre-Generation Screener:** Before user text touches any generative model, the client executes zero-latency regex matching against 60 curated Libyan crisis markers across 360+ colloquial spellings. If ideation or severe collapse is detected, drafting is completely suspended.
- **2. Benign Idiom Exemption:** Colloquial Libyan Arabic frequently uses hyperbolic expressions of death playfully (e.g., 'نموت من الضحك' = dying of laughter, 'نموت فيك' = I love you to death, 'نموت من الجوع' = starving). Exactly 7 benign idioms are stripped prior to screening, eliminating false positives.
- **3. Clinical Blacklist:** Post-generation regex intercepts 55 clinical conditions, psychiatric diagnoses, and medication names. If an LLM hallucinates a diagnosis, the output is instantly blocked and replaced with a safe fallback template.

4.2 The Lethal Hazard of the 'Dead Line' & The contacts: [] Policy

[!] SAFETY RATIONALE: WHY contacts: [] IS AN INTENTIONAL ETHICAL CHOICE

CRITICAL ETHICAL PROTOCOL: In acute suicidal distress, providing an unverified, disconnected, or unstaffed hotline number is lethal. When an individual in active crisis summons the courage to call and hears a dead tone ('الرقم المطلوب غير متاح') or unanswered ringing, feelings of abandonment and acute hopelessness spike, sharply accelerating suicide lethality.

Under our strict safety protocol (safety/contact-verification.md), a telephone number may only be displayed if personally called and verified by a named team member within 8 hours. Because no 24/7 dedicated crisis line in Libya could be verified for continuous responsiveness, contacts: [] is deliberately maintained as an empty array. Instead, the Support Card displays an unalterable, honest Arabic statement:

«لا نستطيع عرض رقم لم نتحقق منه. تحدث مع شخص تثق فيه الآن، وإذا كنت في خطر مباشر توجّه لأقرب قسم طوارئ.»

(We cannot display an unverified number. Speak with someone you trust right now, and if you are in immediate danger, proceed to the nearest emergency department.)

5. AI Core, Prompting & Rejection of Fine-Tuning

5.1 Foundation Model Selection

Jisr utilizes a high-reliability dual-engine cloud strategy backed by local deterministic templates:

- **Primary Foundation Model: Google Gemini 1.5 Flash** — Exceptional multilingual reasoning, native Arabic tokenization economy, and sub-800ms generation latency.
- **High-Speed Resilience Failover: Meta Llama 3.3 70B Versatile** — Hosted on Groq LPU hardware, providing ultra-fast backup generation and open-weights architecture with deterministic JSON mode.
- **Local Deterministic Engine: safety/plain-templates.json** — 35 pre-vetted template permutations ensuring zero user interruption during total connectivity blackouts.

5.2 Why Fine-Tuning Was Explicitly Rejected

A common misconception in hackathons is that 'real AI' requires fine-tuning a small model from scratch. Our engineering team explicitly rejected fine-tuning smaller open-source models (1B–8B parameters) for three decisive reasons:

- **1. Catastrophic Forgetting & Safety Drift:** Fine-tuning smaller models on small, synthetic health datasets frequently impairs general linguistic reasoning, introduces subtle biases, and degrades safety boundaries.
- **2. Deterministic Safety vs. Probabilistic Weights:** Neural networks operating at non-zero temperature are inherently probabilistic. A fine-tuned model cannot guarantee it will never hallucinate a clinical term. In Jisr's Hybrid Architecture, the LLM is restricted to stylistic phrasing, while safety is 100% deterministic and enforced by code before and after inference.
- **3. Device & Hardware Realities:** Running local SLMs on consumer smartphones in Libya causes severe thermal throttling, battery drain, and multi-gigabyte app bundles (>3 GB). Jisr maintains an ultralight footprint (<65 MB APK, <2 KB API payload), fluidly operating on spotty 3G networks.

6. Empirical Safety Benchmarks & Quality Verification

In strict compliance with Codathon evaluation criteria, Jisr measures its safety claims rather than asserting them. The Guardian Layer was benchmarked against the Ai4LY Synthetic Crisis Benchmark (safety/dev-set.json, N=25):

Benchmark Metric	Empirical Score	95% Confidence Interval (Wilson)	Evaluation Scope & Details
Crisis Detection Recall	100.0% (10/10)	72.2% — 100.0%	Detected direct Libyan dialect, severe collapse, MSA ideation, Arabizi, and English crisis.
False-Alarm Rate (Fall-out)	0.0% (0/11)	0.0% — 25.9%	Correctly passed routine exam stress, family strain, money worries, and 7 benign idioms.
Clinical Output Interception	100.0% (55/55)	93.5% — 100.0%	Deterministic post-filter blocked all 55 conditions, diagnostics, medications, and therapy terms.
Cross-Engine Parity	756 / 756 Passed	100.0%	Identical behavior verified between Node.js Guardian engine and TypeScript mobile client.
Unit Test Suite (selftest.mjs)	450 / 450 Passed	100.0%	Validates JSON schemas, diacritic normalization, 105 template variants, and blacklist injection.
Time-to-Outreach (TTO)	< 60 seconds	Observed	Reduces outreach paralysis from >30 minutes of staring at a blank screen to under 1 minute.
First-Sentence Funnel	≥ 65.0%	Design Target	Target completion rate from initial chip

Target	selection to triggering native WhatsApp/SMS share.		
Human Edit Distance	< 30.0%	Design Target	Verifies draft quality: user only needs minor personal adjustments before sending.

[^] **MATHEMATICAL DESIGN RATIONALE: OPTIMIZING FOR RECALL OVER SPECIFICITY**

TUNING DESIGN PHILOSOPHY: The classifier is deliberately tuned toward over-triggering (High Recall / Sensitivity). The cost of a false positive is merely displaying a harmless, supportive card with unalterable human routing. The cost of a false negative is missing a young person in active suicidal crisis. Zero false negatives is our non-negotiable standard.

7. Technology Stack & Repository Architecture

7.1 Core Technology Stack

Layer / Domain	Technology & Framework	Version	Role in Jisr Architecture
Mobile Client	React Native / Expo SDK	RN 0.74.5 / Expo 51	Cross-platform mobile client (com.ai4ly.jisr) with RTL layout & Share.share()
Web Showcase	Next.js (App Router)	16.4.0	Instant browser evaluation with authentic Arabic typography (npm run dev)
AI Microservice	FastAPI / Uvicorn	0.115 / 0.32	Stateless Python backend with Pydantic validation & httpx async clients
Foundation AI	Google Gemini 1.5 Flash	Latest Flash	Sub-800ms 3-tone drafting, free-text restructuring, and dialect grounding
Failover AI	Groq LPU / Llama 3.3 70B	Llama-3.3-70b-versatile	High-throughput fallback engine with deterministic JSON schema output
Local Storage	AsyncStorage	1.23.1	Sandboxed device-only storage for optional recurrence memory; 0 cloud requests
Distribution	EAS Build (APK Profile)	v1.0.0	Standalone sideloadable APK (builds/jisr-v1.0.0.apk, 63.1 MB, SHA-256 verified)
CI / CD	GitHub Actions	v4	Automated APK compilation and verification workflow (.github/workflows/build-apk.yml)

7.2 Repository Organization (The 5 Functional Clusters)

The repository is organized into five clean, decoupled functional clusters:

- **1. Client Showcases:** App.tsx (Root mobile app), src/app/ (Next.js 16 Web Showcase), src/screens/CaptureScreen.tsx, src/components/ (Trust views: BaselineComparison, FaithfulnessView, OutboundPreview, SupportCardModal), src/services/, src/i18n/ar.json.
- **2. AI Core Microservice:** backend/app/main.py, backend/app/routers/ (check-risk, generate-drafts, faithfulness), backend/app/services/ (Gemini/Groq client, PII sanitizer), backend/prompts/.
- **3. Guardian Safety Engine:** safety/crisis-phrases.json (60 phrases, 360+ variants), safety/forbidden-terms.json (55 terms), safety/plain-templates.json (35 templates), safety/support-card.json, safety/dev-set.json.
- **4. Distribution & Artifacts:** builds/jisr-v1.0.0.apk (63.1 MB), docs/PITCH_DECK.pptx, docs/PITCH_DECK.html, docs/COMMITTEE_QA.md, REPORT.md (35 KB technical report), CITATIONS.md.
- **5. QA & Verification Suite:** safety/selftest.mjs (450 tests), safety/evaluate.mjs (recall evaluation), tests/verify-crisis-parity.mjs (756 checks), tests/verify-trust-components.mjs, tests/test-offline-fallback.mjs.

8. Team Engineering Roster & 12-Hour Chronological Progress

8.1 Roles & Engineering Ownership

Member & Role	Codathon Workstream	Primary Engineering Deliverables & Code Ownership
Mohamed Abdel Wadod Thabet (Team Leader)	Workstream 4: Architecture, Trust, Arabic & Docs Lead	• System architecture & technical report (REPORT.md) • Trust components: BaselineComparison.tsx, FaithfulnessView.tsx, OutboundPreview.tsx • Authentic Libyan Arabic UI dictionary (src/i18n/ar.json) • Pitch Deck (PITCH_DECK.pptx/html) & Committee Defense (COMMITTEE_QA.md) • Composite test automation (npm test) & CI build pipeline
Rayan Khalid Aljabo	Workstream 1: Mobile App & Interaction Engineer	• React Native / Expo UI client (com.ai4ly.jisr, App.tsx) • Capture Screen (CaptureScreen.tsx) with 7 RTL stress chips & 5 recipient selectors • Persistent Human Route button ('تكلّم مع حدّ تّوا') • Standalone Android APK packaging (eas.json, builds/jisr-v1.0.0.apk) • Native OS Share Sheet integration (Share.share) & sandboxed AsyncStorage
Muetazballah Qambar	Workstream 2: AI Core & Backend Engineer	• Stateless Python FastAPI microservice (backend/app/main.py) • Endpoints: POST /api/check-risk, /api/generate-drafts, /api/faithfulness • Multi-LLM integration: Gemini 1.5 Flash (sub-800ms) & Groq Llama 3.3 70B • Client-side & server-side regex PII sanitizers (scrubbing +218 numbers/names) • Pydantic v2 validation schemas & CORS middleware

Shaima Abdulsalam Aljelali	Workstream 3: Safety, Guardian & Evidence Engineer	- Guardian Layer crisis phrase lexicon (60 phrases, 360+ variants) - Clinical boundary blacklist (55 terms, safety/forbidden-terms.json) - Deterministic offline plain templates (35 permutations, plain-templates.json) - Static emergency Support Card (safety/support-card.json) & Hour-8 helpline audit - Empirical evaluation harnesses (evaluate.mjs, selftest.mjs) achieving 100% recall
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8.2 Chronological 12-Hour Hackathon Progress Log

Time Window	Workstream 1: Mobile (Rayan Khalid Aljabo)	Workstream 2: Backend (Muetazballah Qambar)	Workstream 3: Safety (Shaima Abdulsalam Aljelali)	Workstream 4: Lead (Mohamed Abdel Wadod Thabet)
Hours 0.0–0.5 (Inception)	Chose React Native / Expo SDK 51. Agreed on API JSON schemas.	Benchmarked Gemini & Groq from Libyan IPs (<800ms latency).	Formulated Guardian Layer architecture & contact audit.	Repo setup, branch protection, authored 00_SHARED_SETUP.md.
Hours 0.5–3.0 (Foundations)	Scaffolded Expo project. Built Capture Screen with 7 RTL chips & mock API.	Built FastAPI app, Pydantic schemas, and drafted 3-tone system prompts.	Curated crisis-phrases.json & forbidden-terms.json. Drafted Support Card.	Authored Libyan Arabic dictionary (ar.json) & individual role guides.
Hours 3.0–6.0 (Core Dev)	Drafting Screen with 3 tones (Gentle, Direct, Formal) & Support Card modal.	Implemented /api/check-risk & PII regex sanitizer (stripping +218 & names).	Expanded crisis list to 60 phrases across Libyan, MSA & Arabizi. 7 benign idioms.	Engineered BaselineComparison.tsx; formulated Wilson 95% CI equations.
Hours 6.0–8.0 (Checkpoint A)	Integrated Share.share() native sheet. Built sandboxed AsyncStorage.	Deployed FastAPI service to staging; added Groq failover; audited zero-logs.	Hour-8 Contact Decision: audited helplines; enacted contacts: [] empty rule.	Engineered OutboundPreview.tsx & FaithfulnessView.tsx; audited zero-retention.
Hours 8.0–10.0 (Checkpoint B)	Swapped mock API for live backend; 2.5s client timeout with offline fallback.	Integrated Shima's 60 crisis phrases & 55 forbidden terms; added /faithfulness.	Built text normalizer (normalize.mjs) & selftest.mjs (450 test cases).	Tested persistent human route & airplane mode fallback; integrated trust views.
Hours 10.0–12.0 (Freeze & QA)	Verified on Android phones; packaged builds/jisr-v1.0.0.apk. Feature freeze.	Hardened error handling, zero-disk logs, JSON serialization. Feature freeze.	Evaluated dev-set.json (100% recall); built verify-crisis-parity.mjs (756 checks).	Created CI workflow (.github/workflows/build-apk.yml); finalized reports & deck.

9. Official Codathon Submission Compliance Matrix

Codathon Deliverable	Required Scope	Repository Artifact & Evidence Location	Status
Scope A: Source Code	Complete application codebase across modern frameworks	- Web: src/app/ (Next.js 16) - Mobile: App.tsx & src/ (Expo SDK 51) - Backend: backend/ (FastAPI / Gemini)	[VERIFIED]
Demo Artifact: Standalone APK	Sideloadable mobile prototype ready for instant device testing	- builds/jisr-v1.0.0.apk (63.1 MB, SHA-256 verified) - Install & verification guide: builds/README.md	[VERIFIED]
Scope B: Documentation	Root-level architectural, setup, operational & usage guide	- Root Master Guide: README.md (23 KB) - Documentation Hub: docs/README.md	[VERIFIED]
Scope C: Technical Report	In-depth Markdown report with methodology, citations & evaluation	REPORT.md (35 KB comprehensive report with Wilson CIs & 12-hour progress log)	[VERIFIED]
Scope D: Presentation	5-minute pitch deck covering problem, solution & societal impact	- PowerPoint: docs/PITCH_DECK.pptx (16:9 Arabic) - HTML: docs/PITCH_DECK.html - Defense: docs/COMMITTEE_QA.md	[VERIFIED]
Section 4: Tool Citations	Explicit attribution for all models, APIs, and libraries used	CITATIONS.md (Gemini, Llama 3.3, Expo, FastAPI, Uvicorn, GitHub Actions)	[VERIFIED]
Safety & Empirical Quality	Measured recall benchmarks & automated test suites	- Master test command: npm test (450 passed) - Crisis Recall: 100.0% (10/10 caught) - Crisis Parity: 756 checks passed	[VERIFIED]
Administrative Items	Team Lead appointment & contestant email disclosure	- Team Leader: Mohamed Abdel Wadod Thabet (abdwadood2000@gmail.com) - Contestants: Rayan Khalid Aljabo, Muetazballah Qambar, Shaima Abdulsalam Aljelali	[VERIFIED]

10. Evaluation Committee Defense & Q&A Playbook

During the 5-minute Q&A following the presentation, the scientific and industry jury will probe the system's boundaries, ethics, and technical depth. Below are the definitive, evidence-backed defense strategies:

[-] Q1: How is this not just another chatbot or wrapper around ChatGPT?

Chatbots aim to maximize screen engagement and simulate human empathy. In mental health, that carries severe ethical hazards of parasocial dependency and dangerous medical drift. Jisr does the exact opposite: it is an authoring companion designed to minimize screen time. The user interacts for under 60 seconds, approves a draft, and sends it via WhatsApp. Jisr ends at a human; chatbots trap the user in artificial dialogue.

[-] Q2: Where does user data go, and how do you protect privacy?

We adhere to three strict zero-retention guarantees: (1) Zero user accounts, zero logins, and zero usage tracking. (2) Pre-transmission PII scrubbing: client-side regex removes Libyan phone numbers (+218 / 091...), emails, and personal names before text leaves the device. (3) Stateless backend: the FastAPI server holds no database and writes zero user text to disk logs. Optional chat history is strictly sandboxed inside on-device AsyncStorage and wiped with 1 tap.

[-] Q3: What if the AI hallucinates, invents feelings, or writes something harmful?

We deploy defense-in-depth: (1) The Guardian Layer intercepts crisis keywords BEFORE drafting—generation never occurs. (2) The Deterministic Output Filter checks every draft against 55 clinical/medication terms, blocking any medical hallucination. (3) Human-in-the-Loop & The Faithfulness View: the AI never sends anything automatically. The user inspects word attribution, edits the text, and must manually press send in their messaging app.

[-] Q4: Does this product diagnose conditions or replace mental health specialists?

No, and we explicitly design against clinical mimicry. Psychiatrists and psychologists in Libya are scarce and should not be burdened by routine exam stress. Jisr mobilizes existing community pillars (siblings, parents, friends). System prompts and code filters strictly forbid condition naming or therapy jargon, and every screen states: 'Jisr is not a doctor and not a diagnostic tool.'

[-] Q5: Why is this solution culturally and technologically feasible in Libya specifically?

(1) Language: Tuned for Libyan Arabic dialect ('حاس روحي بنجنن', 'مضغوط') and Arabizi. (2) Social Norms: Incorporates family and trusted elders as natural support pillars, avoiding clinical stigma. (3) Infrastructure: Distributed as a lightweight standalone APK (builds/jisr-v1.0.0.apk), features an offline template fallback (plain-templates.json) for power/internet blackouts, and interfaces directly with WhatsApp and Messenger.

[-] Q6: Why didn't you train or fine-tune an Arabic LLM on local mental health data?

In safety-critical emotional care, fine-tuning smaller models is an anti-pattern: it causes catastrophic forgetting and stochastic safety drift. A fine-tuned model cannot guarantee 100% adherence to safety. In

our Hybrid Architecture, the LLM is restricted to stylistic phrasing, while safety is 100% deterministic and enforced by code before and after inference. Furthermore, running local models on phones in Libya causes device freezing, thermal throttling, and multi-gigabyte app sizes.

[-] Q7: Where did your crisis data come from, and did you scrape private conversations?

Zero private conversations, chat logs, or vulnerable youths were scraped—doing so violates research ethics and GDPR. Our data consists of three ethical non-personal typologies: (1) Dialect crisis markers hand-curated from native Libyan idioms and WHO guidelines. (2) Clinical boundary blacklist curated from psychiatric terminology. (3) Gold-standard benchmark vectors ethically synthesized by our team.

11. Quick-Start & Operational Evaluation Guide

11.1 Evaluating the Web Showcase (Instant Browser Evaluation)

The Web Showcase provides immediate browser-based evaluation of the complete application flow:

[i] COMMAND: LAUNCH NEXT.JS 16 WEB SHOWCASE

```
cd Jisr
npm install
npm run dev
```

→ Open `http://localhost:3000` in any browser.

→ Select stress chips, pick a recipient, type an optional thought, and inspect generated drafts and Trust Views.

11.2 Sideloading the Standalone Android APK

A pre-compiled production Android APK is committed directly in the repository for instant physical device testing:

- Location: `builds/jisr-v1.0.0.apk` (63.1 MB, SHA-256 verified)
- Instructions: Transfer to any Android phone (API 23+ / Android 6.0–14), enable 'Install from Unknown Sources', and launch.
- Offline Test: Put the phone in Airplane Mode. Jisr will automatically switch to the local deterministic template engine with zero errors.

11.3 Running the Automated QA Verification Suite

Execute the full automated regression and safety test suite with a single command:

[*] COMMAND: MASTER VERIFICATION SUITE

npm test

Executes in series:

1. node safety/selftest.mjs → 450 unit tests pass (schemas, diacritics, templates, filters)
2. node safety/evaluate.mjs → 100.0% Recall benchmark on dev-set.json (10/10 caught)
3. node tests/verify-trust-components.mjs → Validates RTL Arabic keys and Trust Views
4. node tests/verify-crisis-parity.mjs → 756 parity checks pass between mobile TypeScript & Node engine

12. Conclusion & Future Roadmap

Jisr demonstrates that artificial intelligence in mental health is most transformative when it acts not as an artificial authority or conversational substitute, but as an empowering, frictionless bridge. By giving young Libyans the words to start the conversation, Jisr ends the silence early and returns care to where it belongs: in human hands.

Future Development Horizons:

- **On-Device SLM Inference:** Exploring quantized on-device small language models (SLMs) for 100% offline generation without compromising thermal or battery life.
- **Institutional Partnerships:** Formalizing verification protocols with certified Libyan psychosocial entities and community health centers.
- **Expanded Recipient Registers:** Adding customized communication registers for university academic advisors and workplace supervisors.